Parallel beta-eta reduction contracts toward the beta-eta-complete development: if M beredpar M' then M' beredpar becd(M)

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the delegation names the source lemma and the eta case is expanded verbatim including its side condition x not in FV(N); the closing 'immediate by induction hypothesis' identifies the exact hypothesis used.

Editorial note. The label lem:cont names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Editorial note. OLP reuses this label inside the unit; the card is identified by its statement span (source lines 730-732) and the claim key carries the statement start line.

Allowed standard prerequisites (editorial).

- Defining equations of the beta-eta complete development. Same unit, Definition defn:becd with equations \ollabel{defn:becd1}-\ollabel{defn:becd5}. Each case rewrites becd of a compound term; equation becd5 handles the eta contraction case.
- Substitution compatibility for the eta-parallel relation. Same unit, \ollabel{lem:comp} (source line 712). Redex case: transports the pair of induction hypotheses through substitution.
- The beta-section derivation induction. Same unit, \ollabel{lem:cont} of the parallel-beta-reduction section (source line 857). The first four cases are declared 'like those in' that proof; the fifth, eta, rule is the new content.

Source reference. Original definition defn-beredpar5: sources/olp-lambda-calculus-church-rosser/pinned-original-source.tex, source lines 680–680. Prerequisite pointer

invoked at proof line(s) 738; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If $M \to_\beta M'$ then $M' \to_\beta \ulcorner M$.*

Proof. By induction on the !!derivation of $M \to_\beta M'$.

The first four cases are like those in 0.1. If the last rule is defn-beredpar5, then M is $\lambda[x][Nx]$ and M' is N' for some x, N, N' where $x \notin FV(N)$ and $N \to_\beta N'$. We want to show that $N' \to_\beta \ulcorner(\lambda[x][Nx])$, i.e., $N' \to_\beta \ulcorner N$, which is immediate by induction hypothesis. □